

Abhuday Mishra

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Summary

Backend Engineer with 3+ years of experience building distributed and AI-powered systems at scale. Specialized in **multi-tenant data platforms**, **RAG/LLM infrastructure**, and production observability, with experience owning end-to-end backend systems across **ingestion, retrieval, evaluation, and deployment workflows**.

SKILLS SUMMARY

- **Core Engineering:** Java, Spring Boot, Python, FastAPI, Distributed Systems, Microservices, REST APIs
- **Cloud & Infrastructure:** AWS (Lambda, SQS, S3, EKS, Fargate), Docker, Kubernetes
- **Data & Storage:** PostgreSQL, Cassandra, Redis, DynamoDB, MongoDB
- **System Design:** Scalable Systems, Event-Driven Architecture, Fault Tolerance, High Availability
- **AI / Data Systems:** RAG Pipelines, Prompt Engineering, Vector Search, AWS Bedrock, LLM Evaluation
- **Observability & Tooling:** Structured Logging, Cost Attribution, Anomaly Detection, CI/CD Pipelines

EXPERIENCE

- **AiDash** Apr 2025 - Present
 - *Software Engineer 2*
 - **WMP (Wildfire Mitigation Plan)**
 - * Designed and deployed a **multi-tenant RAG pipeline** enabling utility companies to query wildfire mitigation documents, supporting **100+ tenants** across a **polyglot microservices architecture** (Java + Python)
 - * Built an **LLM evaluation framework** with a categorized golden test set; measured retrieval (**Precision@K**, **Recall@K**, **MRR**) and generation metrics (**Faithfulness**, **Hallucination rate**) separately across **7 question categories**
 - * Identified retrieval as the primary bottleneck (avg **3.06/5** vs generation **3.77/5**); used **per-category breakdown** to drive architectural decisions including **parser migration** and **dynamic top-k routing**
 - * Solved **multi-tenant vector isolation** under **AWS Bedrock** constraints using **metadata-based filtering** with server-validated tenant identity, avoiding per-tenant KB duplication across **100+ tenants**
 - * Optimized cold-start latency to **<2s** by constraining AI services to **128MB pods**; reduced inference cost to **\$0.019/query** through chunking strategy, retrieval depth, and model selection tuning
 - * Implemented **structured observability** across the RAG pipeline — logging every query, retrieval result, token count, and guardrail trigger — enabling **cost attribution per tenant** and **anomaly detection**
 - **CRIS 2.0 (Climate Risk Intelligence System)**
 - * Architected and built a **distributed ingestion pipeline** powering CRIS Premium & Essentials, enabling onboarding of **20+ enterprise tenants within 3 months**
 - * Eliminated duplicate ingestion across **concurrent scheduler pods** by implementing **idempotent processing** and coordination mechanisms, improving data consistency
 - * Re-architected a blocking synchronous pipeline into an **event-driven async system**, increasing throughput to **320k+ records/hour**
 - * Designed a **config-driven onboarding** framework, reducing new tenant setup effort to near-zero and eliminating code changes for new data models
- **AiDash** Jul 2023 - Mar 2025
 - *Software Engineer*
 - **Serverless Data Archival System**
 - * Designed and built a **serverless data archival pipeline** using AWS Fargate, SQS, Lambda, S3, and DynamoDB to offload cold data from RDS
 - * Reduced primary RDS storage from **400GB to 3GB**, significantly improving query performance and lowering infrastructure cost
 - * Enabled seamless access to archived data via **DynamoDB and S3**, maintaining low-latency retrieval for historical records
 - **Multi-Tenant Data Ingestion System**
 - * Designed a **multi-tenant ingestion system** processing **300k+ records** every 6 hours within AWS Lambda constraints
 - * Optimized memory usage via **chunked processing (1k records/batch)**, reducing peak memory from **138MB to 10MB (93%)**
 - * Reduced in-memory object load from **80k to 5k**, improving execution stability under high concurrency
 - * Eliminated need for higher Lambda memory tiers, reducing infrastructure cost by **20%**

EDUCATION

- **SRM Institute Of Science and Technology** Chennai, India
 - *Bachelor of Technology - Computer Science and Engineering; GPA: 9.17* Jun 2019 - May 2023

PROJECTS

- **RCA Analyzer — Python, FastAPI, React, MCP, AWS:** Built an AI-powered RCA analyzer leveraging LLMs to infer root causes from distributed logs; awarded 2nd place in a company-wide hackathon